CS 4265 Big Data Analytics
Milestone 2 Progress Report

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Project Overview

The goal of this project is to build a distributed analytics pipeline using Apache Spark. For Milestone 2, the objective is to demonstrate that the basic components of the pipeline work, including data loading, distributed processing, and pipeline structure.

The project currently uses a Steam recommendation dataset that contains more than **41 million user interaction records**. This dataset is significantly larger than the dataset originally proposed and is more suitable for demonstrating Big Data processing.

Current Progress

So far, the basic proof-of-concept pipeline has been implemented and tested locally using Apache Spark.

First, the Spark environment was successfully initialized and configured to run locally. The pipeline was able to load the dataset from CSV files and process the data using Spark DataFrames.

Second, the system successfully loaded two large datasets: the game metadata dataset and the recommendation dataset. The pipeline processed approximately **50,872 game records** and **41,154,794 recommendation records**. This confirms that the system can handle datasets large enough for a Big Data project.

Third, a distributed join operation was executed between the recommendation dataset and the game dataset using the `app_id` column. The joined results were displayed and the schema of the dataset was successfully generated.

Overall, these results show that the main components of the pipeline are working correctly.

Challenges Encountered

One challenge encountered during testing was related to writing the output dataset in Parquet format on the Windows environment. Although the Spark pipeline successfully exe-

cuted the data loading and join operations, the output stage failed due to a missing Hadoop configuration variable (`HADOOP_HOME`).

This issue is related to the local system configuration rather than the pipeline logic itself. Since the processing steps were completed successfully before the output stage, the proof-of-concept pipeline is still considered functional.

Changes from Milestone 1

The main change from Milestone 1 was the dataset used in the project.

The original proposal used a stock market dataset, but the dataset size was relatively small and did not fully represent a Big Data workload. To address this issue, the project pivoted to a Steam recommendation dataset containing over 41 million records.

This new dataset better demonstrates distributed data processing and large-scale analytics.

Next Steps

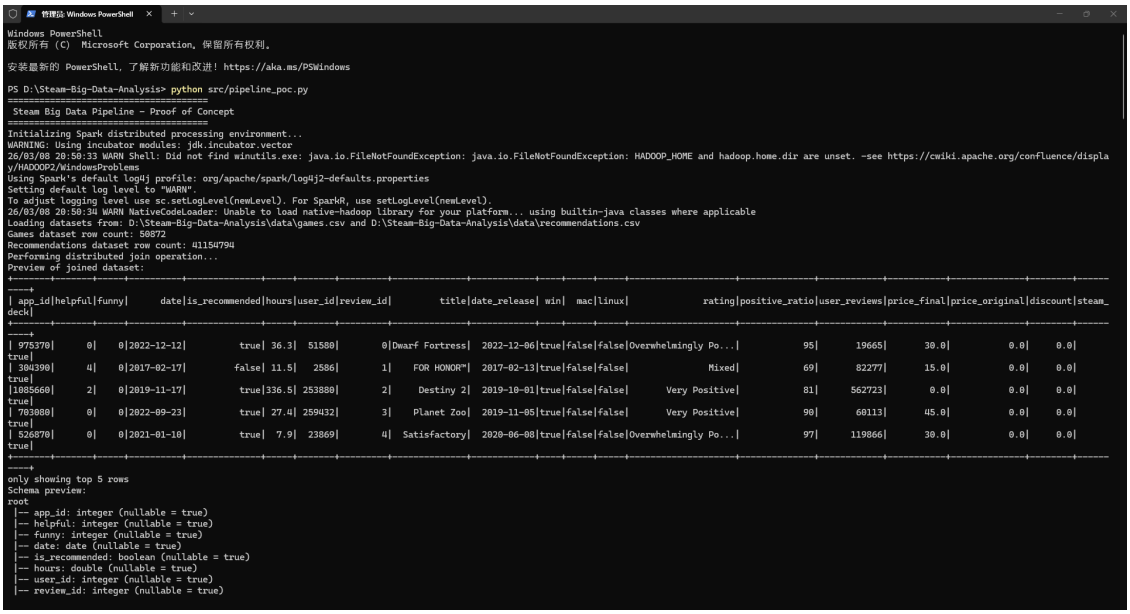
The next step of the project is to improve the pipeline and complete the storage stage.

Planned tasks include:

- resolving the Parquet output issue
- loading additional datasets such as the user dataset
- performing distributed aggregations
- running analytical queries using Spark SQL

Pipeline Execution Evidence

The following screenshots show the successful execution of the Spark pipeline.

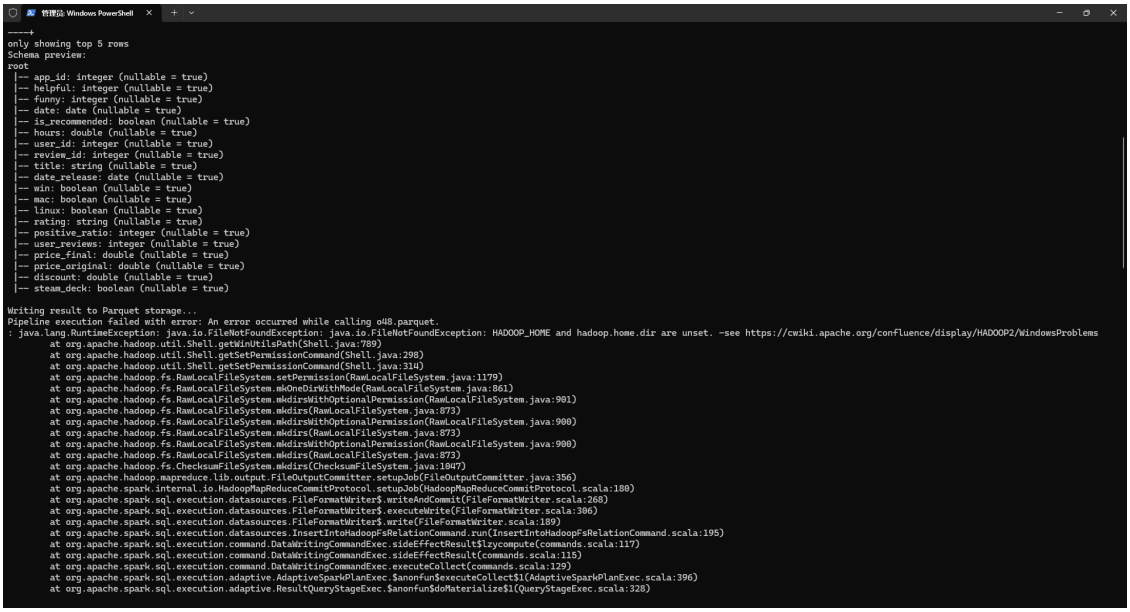


```
管理员: Windows PowerShell
Windows PowerShell
版权所有 (C)  Microsoft Corporation。保留所有权利。

安装最新的 PowerShell，了解新功能和改进！ https://aka.ms/PSWindows

PS D:\Steam-Big-Data-Analysis> python src/pipeline_poc.py
====================================
 Steam Big Data Pipeline - Proof of Concept
====================================
Initializing Spark distributed processing environment...
WARNING: Using incubator modules: jdk.incubator.vector
26/03/08 20:50:33 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/displa
y/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/08 20:50:34 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Loading datasets from: D:\Steam-Big-Data-Analysis\data\games.csv and D:\Steam-Big-Data-Analysis\data\recommendations.csv
Games dataset row count: 50872
Recommendations dataset row count: 41154794
Performing distributed join operation...
Preview of joined dataset:
+-------+-------+-----+----------+--------------+-----+-------+---------+-------------+------------+----+-----+-----+--------------------+-------------+------------+-----------+-------------+--------+------
----+
| app_id|helpful|funny|      date|is_recommended|hours|user_id|review_id|        title|date_release| win|  mac|linux|             rating|positive_ratio|user_reviews|price_final|price_original|discount|steam_
deck|
+-------+-------+-----+----------+--------------+-----+-------+---------+-------------+------------+----+-----+-----+--------------------+-------------+------------+-----------+-------------+--------+------
----+
| 975370|      0|    0|2022-12-12|          true| 36.3|  51580|        0|Dwarf Fortress|  2022-12-06|true|false|false|Overwhelmingly Po...|           95|       19665|       30.0|          0.0|     0.0|
true|
| 394390|      4|    0|2017-02-17|         false| 11.5|   2586|        1|    FOR HONOR™|  2017-02-13|true|false|false|               Mixed|           69|       82277|       15.0|          0.0|     0.0|
true|
|1085660|      2|    0|2019-11-17|          true|336.5| 253880|        2|     Destiny 2|  2019-10-01|true|false|false|       Very Positive|           81|      562723|        0.0|          0.0|     0.0|
true|
| 703080|      0|    0|2022-09-23|          true| 27.4| 259432|        3|    Planet Zoo|  2019-11-05|true|false|false|       Very Positive|           90|       60113|       45.0|          0.0|     0.0|
true|
| 526870|      0|    0|2021-01-10|          true|  7.9|  23869|        4|  Satisfactory|  2020-06-08|true|false|false|Overwhelmingly Po...|           97|      119866|       30.0|          0.0|     0.0|
true|
+-------+-------+-----+----------+--------------+-----+-------+---------+-------------+------------+----+-----+-----+--------------------+-------------+------------+-----------+-------------+--------+------
----+
only showing top 5 rows
Schema preview:
root
 |-- app_id: integer (nullable = true)
 |-- helpful: integer (nullable = true)
 |-- funny: integer (nullable = true)
 |-- date: date (nullable = true)
 |-- is_recommended: boolean (nullable = true)
 |-- hours: double (nullable = true)
 |-- user_id: integer (nullable = true)
 |-- review_id: integer (nullable = true)
```

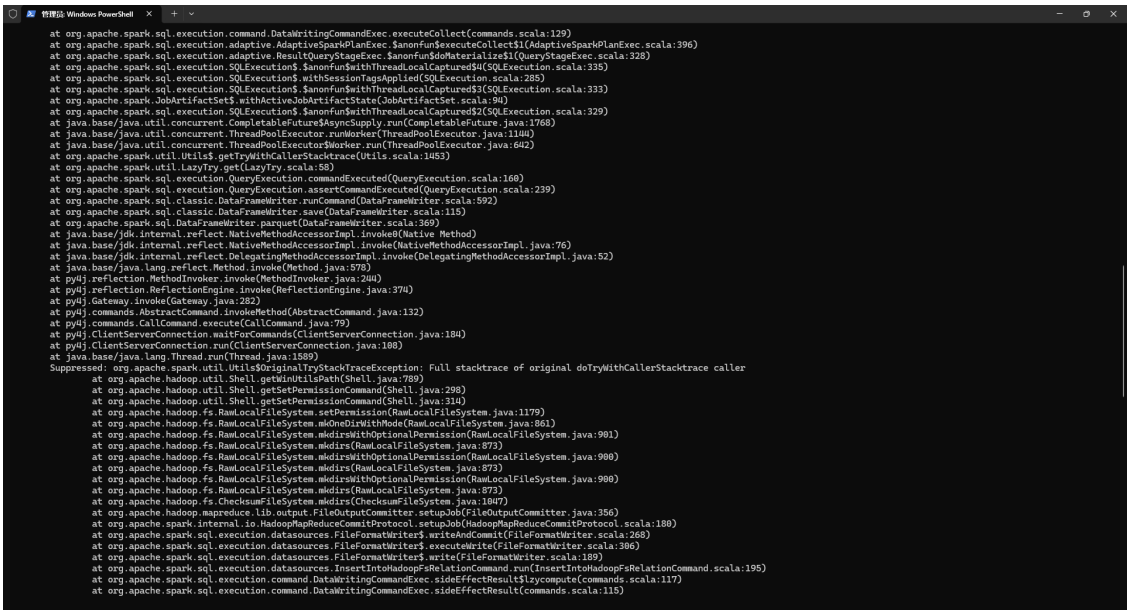
Figure 1: Spark environment initialization



```
管理员: Windows PowerShell
----+
only showing top 5 rows
Schema preview:
root
 |-- app_id: integer (nullable = true)
 |-- helpful: integer (nullable = true)
 |-- funny: integer (nullable = true)
 |-- date: date (nullable = true)
 |-- is_recommended: boolean (nullable = true)
 |-- hours: double (nullable = true)
 |-- user_id: integer (nullable = true)
 |-- review_id: integer (nullable = true)
 |-- title: string (nullable = true)
 |-- date_release: date (nullable = true)
 |-- win: boolean (nullable = true)
 |-- mac: boolean (nullable = true)
 |-- linux: boolean (nullable = true)
 |-- rating: string (nullable = true)
 |-- positive_ratio: integer (nullable = true)
 |-- user_reviews: integer (nullable = true)
 |-- price_final: double (nullable = true)
 |-- price_original: double (nullable = true)
 |-- discount: double (nullable = true)
 |-- steam_deck: boolean (nullable = true)

Writing result to Parquet storage...
Pipeline execution failed with error: An error occurred while calling o48.parquet.
: java.lang.RuntimeException: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
        at org.apache.hadoop.util.Shell.getWinUtilsPath(Shell.java:799)
        at org.apache.hadoop.util.Shell.getSetPermissionCommand(Shell.java:298)
        at org.apache.hadoop.util.Shell.getSetPermissionCommand(Shell.java:314)
        at org.apache.hadoop.fs.RawLocalFileSystem.setPermission(RawLocalFileSystem.java:1179)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkOneDirWithMode(RawLocalFileSystem.java:861)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkdirsWithOptionalPermission(RawLocalFileSystem.java:901)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkdirs(RawLocalFileSystem.java:873)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkdirsWithOptionalPermission(RawLocalFileSystem.java:900)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkdirs(RawLocalFileSystem.java:873)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkdirsWithOptionalPermission(RawLocalFileSystem.java:900)
        at org.apache.hadoop.fs.RawLocalFileSystem.mkdirs(RawLocalFileSystem.java:873)
        at org.apache.hadoop.fs.ChecksumFileSystem.mkdirs(ChecksumFileSystem.java:1047)
        at org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter.setupJob(FileOutputCommitter.java:356)
        at org.apache.spark.internal.io.HadoopMapReduceCommitProtocol.setupJob(HadoopMapReduceCommitProtocol.scala:180)
        at org.apache.spark.sql.execution.datasources.FileFormatWriter$.writeAndCommit(FileFormatWriter.scala:268)
        at org.apache.spark.sql.execution.datasources.FileFormatWriter$.executeWrite(FileFormatWriter.scala:306)
        at org.apache.spark.sql.execution.datasources.FileFormatWriter$.write(FileFormatWriter.scala:189)
        at org.apache.spark.sql.execution.datasources.InsertIntoHadoopFsRelationCommand.run(InsertIntoHadoopFsRelationCommand.scala:195)
        at org.apache.spark.sql.execution.command.DataWritingCommandExec.sideEffectResult$lzycompute(commands.scala:117)
        at org.apache.spark.sql.execution.command.DataWritingCommandExec.sideEffectResult(commands.scala:115)
        at org.apache.spark.sql.execution.command.DataWritingCommandExec.executeCollect(commands.scala:129)
        at org.apache.spark.sql.execution.adaptive.AdaptiveSparkPlanExec.$anonfun$executeCollect$1(AdaptiveSparkPlanExec.scala:396)
        at org.apache.spark.sql.execution.adaptive.ResultQueryStageExec.$anonfun$doMaterialize$1(QueryStageExec.scala:328)
```

Figure 2: Dataset loading and row counts



```
管理员: Windows PowerShell
        at org.apache.spark.sql.execution.command.DataWritingCommandExec.executeCollect(commands.scala:129)
        at org.apache.spark.sql.execution.adaptive.AdaptiveSparkPlanExec.$anonfun$executeCollect$1(AdaptiveSparkPlanExec.scala:396)
        at org.apache.spark.sql.execution.adaptive.ResultQueryStageExec.$anonfun$doMaterialize$1(QueryStageExec.scala:328)
        at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$4(SQLExecution.scala:335)
        at org.apache.spark.sql.execution.SQLExecution$.withSessionTagsApplied(SQLExecution.scala:285)
        at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$3(SQLExecution.scala:333)
        at org.apache.spark.JobArtifactSet$.withActiveJobArtifactState(JobArtifactSet.scala:94)
        at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$2(SQLExecution.scala:329)
        at java.base/java.util.concurrent.CompletableFuture$AsyncSupply.run(CompletableFuture.java:1768)
        at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1144)
        at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:642)
        at org.apache.spark.util.Utils$.getTryWithCallerStacktrace(Utils.scala:1453)
        at org.apache.spark.util.LazyTry.get(LazyTry.scala:58)
        at org.apache.spark.sql.execution.QueryExecution.commandExecuted(QueryExecution.scala:160)
        at org.apache.spark.sql.execution.QueryExecution.assertCommandExecuted(QueryExecution.scala:239)
        at org.apache.spark.sql.classic.DataFrameWriter.runCommand(DataFrameWriter.scala:592)
        at org.apache.spark.sql.classic.DataFrameWriter.save(DataFrameWriter.scala:115)
        at org.apache.spark.sql.DataFrameWriter.parquet(DataFrameWriter.scala:369)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:76)
        at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:52)
        at java.base/java.lang.reflect.Method.invoke(Method.java:578)
        at py4j.reflection.MethodInvoker.invoke(MethodInvoker.java:244)
        at py4j.reflection.ReflectionEngine.invoke(ReflectionEngine.java:374)
        at py4j.Gateway.invoke(Gateway.java:282)
        at py4j.commands.AbstractCommand.invokeMethod(AbstractCommand.java:132)
        at py4j.commands.CallCommand.execute(CallCommand.java:79)
        at py4j.ClientServerConnection.waitForCommands(ClientServerConnection.java:184)
        at py4j.ClientServerConnection.run(ClientServerConnection.java:108)
        at java.base/java.lang.Thread.run(Thread.java:1589)
        Suppressed: org.apache.spark.util.Utils$OriginalTryStackTraceException: Full stacktrace of original doTryWithCallerStacktrace caller
                at org.apache.hadoop.util.Shell.getWinUtilsPath(Shell.java:789)
                at org.apache.hadoop.util.Shell.getSetPermissionCommand(Shell.java:298)
                at org.apache.hadoop.util.Shell.getSetPermissionCommand(Shell.java:314)
                at org.apache.hadoop.fs.RawLocalFileSystem.setPermission(RawLocalFileSystem.java:1179)
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                at org.apache.spark.sql.execution.command.DataWritingCommandExec.sideEffectResult$lzycompute(commands.scala:117)
                at org.apache.spark.sql.execution.command.DataWritingCommandExec.sideEffectResult(commands.scala:115)
```
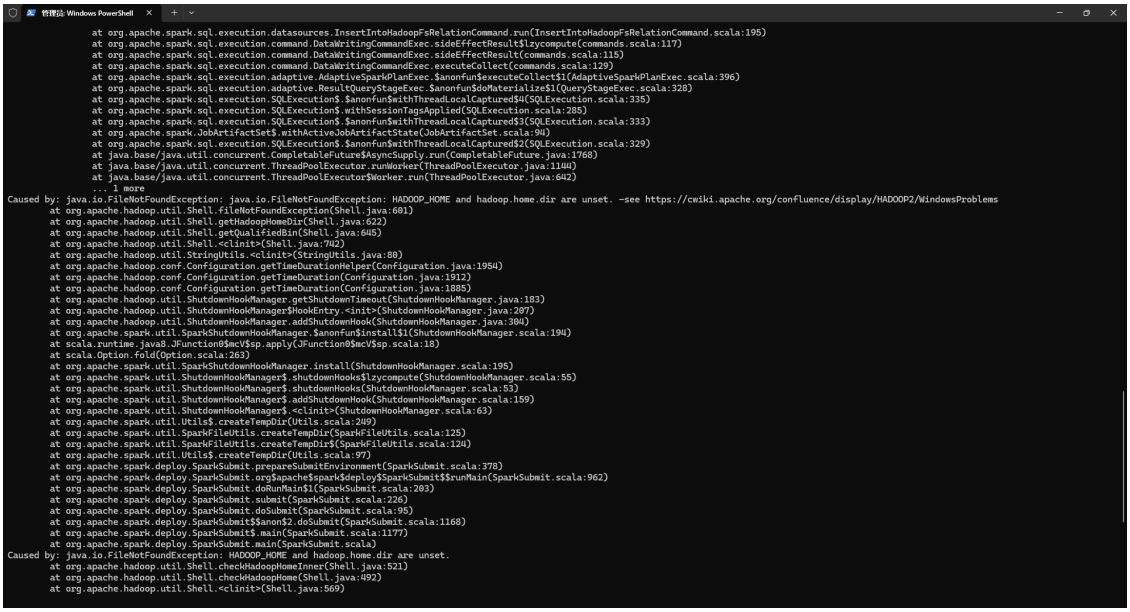
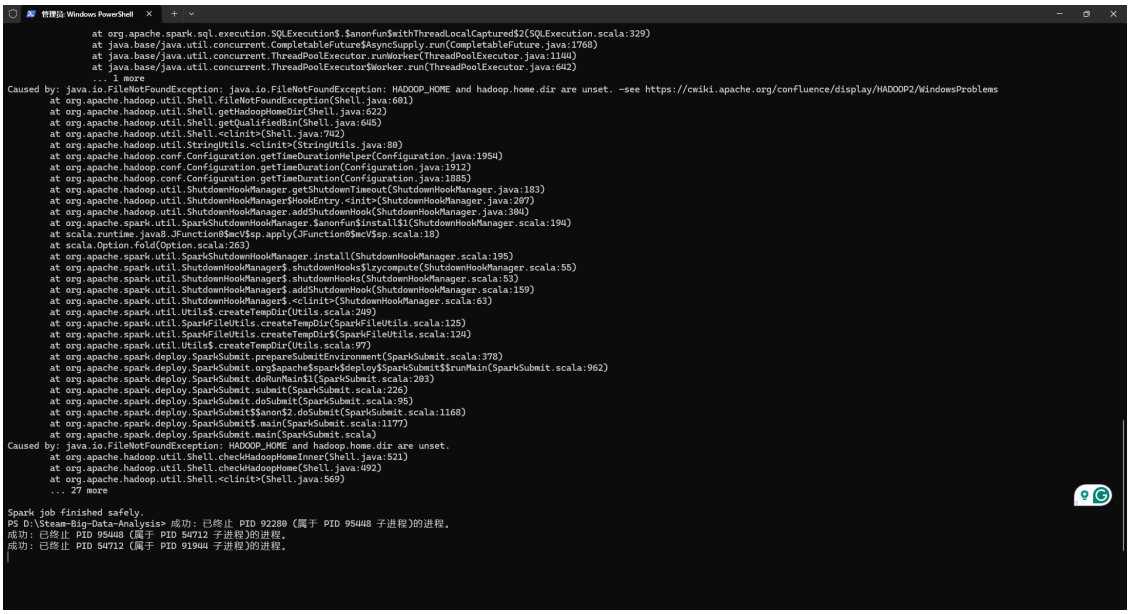

Figure 3: Preview of joined dataset



```
管理员: Windows PowerShell
                at org.apache.spark.sql.execution.datasources.InsertIntoHadoopFsRelationCommand.run(InsertIntoHadoopFsRelationCommand.scala:195)
                at org.apache.spark.sql.execution.command.DataWritingCommandExec.sideEffectResult$lzycompute(commands.scala:117)
                at org.apache.spark.sql.execution.command.DataWritingCommandExec.sideEffectResult(commands.scala:115)
                at org.apache.spark.sql.execution.command.DataWritingCommandExec.executeCollect(commands.scala:129)
                at org.apache.spark.sql.execution.adaptive.AdaptiveSparkPlanExec.$anonfun$executeCollect$1(AdaptiveSparkPlanExec.scala:396)
                at org.apache.spark.sql.execution.adaptive.ResultQueryStageExec.$anonfun$doMaterialize$1(QueryStageExec.scala:328)
                at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$4(SQLExecution.scala:335)
                at org.apache.spark.sql.execution.SQLExecution$.withSessionTagsApplied(SQLExecution.scala:285)
                at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$3(SQLExecution.scala:333)
                at org.apache.spark.JobArtifactSet$.withActiveJobArtifactState(JobArtifactSet.scala:94)
                at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$2(SQLExecution.scala:329)
                at java.base/java.util.concurrent.CompletableFuture$AsyncSupply.run(CompletableFuture.java:1768)
                at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1144)
                at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:642)
                ... 1 more
Caused by: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
        at org.apache.hadoop.util.Shell.fileNotFoundException(Shell.java:601)
        at org.apache.hadoop.util.Shell.getHadoopHomeDir(Shell.java:622)
        at org.apache.hadoop.util.Shell.getQualifiedBin(Shell.java:645)
        at org.apache.hadoop.util.Shell.<clinit>(Shell.java:742)
        at org.apache.hadoop.util.StringUtils.<clinit>(StringUtils.java:80)
        at org.apache.hadoop.conf.Configuration.getTimeDurationHelper(Configuration.java:1954)
        at org.apache.hadoop.conf.Configuration.getTimeDuration(Configuration.java:1912)
        at org.apache.hadoop.conf.Configuration.getTimeDuration(Configuration.java:1885)
        at org.apache.hadoop.util.ShutdownHookManager.getShutdownTimeout(ShutdownHookManager.java:183)
        at org.apache.hadoop.util.ShutdownHookManager$HookEntry.<init>(ShutdownHookManager.java:207)
        at org.apache.hadoop.util.ShutdownHookManager.addShutdownHook(ShutdownHookManager.java:304)
        at org.apache.spark.util.SparkShutdownHookManager.$anonfun$install$1(ShutdownHookManager.scala:194)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.scala:18)
        at scala.Option.fold(Option.scala:263)
        at org.apache.spark.util.SparkShutdownHookManager.install(ShutdownHookManager.scala:195)
        at org.apache.spark.util.ShutdownHookManager$.shutdownHooks$lzycompute(ShutdownHookManager.scala:55)
        at org.apache.spark.util.ShutdownHookManager$.shutdownHooks(ShutdownHookManager.scala:53)
        at org.apache.spark.util.ShutdownHookManager$.addShutdownHook(ShutdownHookManager.scala:159)
        at org.apache.spark.util.ShutdownHookManager$.<clinit>(ShutdownHookManager.scala:63)
        at org.apache.spark.util.Utils$.createTempDir(Utils.scala:249)
        at org.apache.spark.util.SparkFileUtils.createTempDir(SparkFileUtils.scala:125)
        at org.apache.spark.util.SparkFileUtils.createTempDir$(SparkFileUtils.scala:124)
        at org.apache.spark.util.Utils$.createTempDir(Utils.scala:97)
        at org.apache.spark.deploy.SparkSubmit.prepareSubmitEnvironment(SparkSubmit.scala:378)
        at org.apache.spark.deploy.SparkSubmit.org$apache$spark$deploy$SparkSubmit$$runMain(SparkSubmit.scala:962)
        at org.apache.spark.deploy.SparkSubmit.doRunMain$1(SparkSubmit.scala:203)
        at org.apache.spark.deploy.SparkSubmit.submit(SparkSubmit.scala:226)
        at org.apache.spark.deploy.SparkSubmit.doSubmit(SparkSubmit.scala:95)
        at org.apache.spark.deploy.SparkSubmit$$anon$2.doSubmit(SparkSubmit.scala:1168)
        at org.apache.spark.deploy.SparkSubmit$.main(SparkSubmit.scala:1177)
        at org.apache.spark.deploy.SparkSubmit.main(SparkSubmit.scala)
Caused by: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset.
        at org.apache.hadoop.util.Shell.checkHadoopHomeInner(Shell.java:521)
        at org.apache.hadoop.util.Shell.checkHadoopHome(Shell.java:492)
        at org.apache.hadoop.util.Shell.<clinit>(Shell.java:569)
```

Figure 4: Schema preview



```
管理员: Windows PowerShell
                at org.apache.spark.sql.execution.SQLExecution$.$anonfun$withThreadLocalCaptured$2(SQLExecution.scala:329)
                at java.base/java.util.concurrent.CompletableFuture$AsyncSupply.run(CompletableFuture.java:1768)
                at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1144)
                at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:642)
                ... 1 more
Caused by: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
        at org.apache.hadoop.util.Shell.fileNotFoundException(Shell.java:601)
        at org.apache.hadoop.util.Shell.getHadoopHomeDir(Shell.java:622)
        at org.apache.hadoop.util.Shell.getQualifiedBin(Shell.java:645)
        at org.apache.hadoop.util.Shell.<clinit>(Shell.java:742)
        at org.apache.hadoop.util.StringUtils.<clinit>(StringUtils.java:80)
        at org.apache.hadoop.conf.Configuration.getTimeDurationHelper(Configuration.java:1954)
        at org.apache.hadoop.conf.Configuration.getTimeDuration(Configuration.java:1912)
        at org.apache.hadoop.conf.Configuration.getTimeDuration(Configuration.java:1885)
        at org.apache.hadoop.util.ShutdownHookManager.getShutdownTimeout(ShutdownHookManager.java:183)
        at org.apache.hadoop.util.ShutdownHookManager$HookEntry.<init>(ShutdownHookManager.java:207)
        at org.apache.hadoop.util.ShutdownHookManager.addShutdownHook(ShutdownHookManager.java:304)
        at org.apache.spark.util.SparkShutdownHookManager.$anonfun$install$1(ShutdownHookManager.scala:194)
        at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.scala:18)
        at scala.Option.fold(Option.scala:263)
        at org.apache.spark.util.SparkShutdownHookManager.install(ShutdownHookManager.scala:195)
        at org.apache.spark.util.ShutdownHookManager$.shutdownHooks$lzycompute(ShutdownHookManager.scala:55)
        at org.apache.spark.util.ShutdownHookManager$.shutdownHooks(ShutdownHookManager.scala:53)
        at org.apache.spark.util.ShutdownHookManager$.addShutdownHook(ShutdownHookManager.scala:159)
        at org.apache.spark.util.ShutdownHookManager$.<clinit>(ShutdownHookManager.scala:63)
        at org.apache.spark.util.Utils$.createTempDir(Utils.scala:249)
        at org.apache.spark.util.SparkFileUtils.createTempDir(SparkFileUtils.scala:125)
        at org.apache.spark.util.SparkFileUtils.createTempDir$(SparkFileUtils.scala:124)
        at org.apache.spark.util.Utils$.createTempDir(Utils.scala:97)
        at org.apache.spark.deploy.SparkSubmit.prepareSubmitEnvironment(SparkSubmit.scala:378)
        at org.apache.spark.deploy.SparkSubmit.org$apache$spark$deploy$SparkSubmit$$runMain(SparkSubmit.scala:962)
        at org.apache.spark.deploy.SparkSubmit.doRunMain$1(SparkSubmit.scala:203)
        at org.apache.spark.deploy.SparkSubmit.submit(SparkSubmit.scala:226)
        at org.apache.spark.deploy.SparkSubmit.doSubmit(SparkSubmit.scala:95)
        at org.apache.spark.deploy.SparkSubmit$$anon$2.doSubmit(SparkSubmit.scala:1168)
        at org.apache.spark.deploy.SparkSubmit$.main(SparkSubmit.scala:1177)
        at org.apache.spark.deploy.SparkSubmit.main(SparkSubmit.scala)
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        at org.apache.hadoop.util.Shell.checkHadoopHomeInner(Shell.java:521)
        at org.apache.hadoop.util.Shell.checkHadoopHome(Shell.java:492)
        at org.apache.hadoop.util.Shell.<clinit>(Shell.java:569)
        ... 27 more

Spark job finished safely.
PS D:\Steam-Big-Data-Analysis> 成功: 已终止 PID 92280 (属于 PID 95448 子进程)的进程。
成功: 已终止 PID 95448 (属于 PID 54712 子进程)的进程。
成功: 已终止 PID 54712 (属于 PID 91944 子进程)的进程。
```

Figure 5: Output stage error message